

Synthesis (Zero-RISC)

Ahmed Seddik - Mohamed Abdelfattah - Mustafa Hendawi

Outlines

- Ibex configurations
- Challenges
- Effective commands
- Solution
- Results
- Inquiries
- Appendix
 - Synthesis script
 - Formality script

Ibex Configurations

Config	"micro"	"small"	"maxperf"	"maxperf-pmp-bmfull"
Features	RV32EC	RV32IMC, 3 cycle mult	RV32IMC, 1 cycle mult, Branch target ALU, Writeback stage	RV32IMCB, 1 cycle mult, Branch target ALU, Writeback stage, 16 PMP regions
Performance (CoreMark/MHz)	0.904	2.47	3.13	3.13
Area - Yosys (kGE)	16.85	26.60	32.48	66.02
Area - Commercial (estimated kGE)	~15	~24	~30	~61
Verification status	Red	Green	Green	Green

Small (default)

Ibex configurations

```
# Two-stage pipeline without additional branch target ALU and 3 cycle multiplier
# (4 cycles for mulh), resulting in 2 stall cycles for mul (3 for mulh)
small:
  RV32E                : 0
  RV32M                : "ibex_pkg::RV32MFast"
  RV32B                : "ibex_pkg::RV32BNone"
  RegFile              : "ibex_pkg::RegFileFF"
  BranchTargetALU      : 0
  WritebackStage        : 0
  ICache               : 0
  ICacheECC            : 0
  ICacheScramble        : 0
  BranchPredictor      : 0
  DbgTriggerEn         : 0
  SecureIbex           : 0
  PMPEnable            : 0
  PMPGranularity        : 0
  PMPNumRegions         : 4
  MHPMCounterNum       : 0
  MHPMCounterWidth     : 40
```

Max.Perf.

Ibex configurations

```
# Three-stage pipeline with additional branch target ALU and 1 cycle multiplier
# (2 cycles for mulh) so mul does not stall (mulh stall 1 cycles). This is the
# maximum performance configuration ignoring the branch predictor (which isn't
# yet fully verified)
maxperf:
  RV32E           : 0
  RV32M           : "ibex_pkg::RV32MSingleCycle"
  RV32B           : "ibex_pkg::RV32BNone"
  RegFile         : "ibex_pkg::RegFileFF"
  BranchTargetALU : 1
  WritebackStage  : 1
  ICache          : 0
  ICacheECC       : 0
  ICacheScramble  : 0
  BranchPredictor : 0
  DbgTriggerEn    : 0
  SecureIbex      : 0
  PMPEnable       : 0
  PMPGranularity  : 0
  PMPNumRegions   : 4
  MHPMCounterNum  : 0
  MHPMCounterWidth : 40
```

Challenges

- RTL wasn't ready for ASIC tools.
- Cell count is very small.
- Unable to resolve designs in the link phase.
- Setup violation
- Unmatched points in Formality check.

RTL wasn't ready for ASIC tools

Challenges

- Analyzed the hierarchy using Vivado.



RTL wasn't ready for ASIC tools

Challenges

- Assertions were used with default values.

```
// Macro bodies included by prim_assert.sv for tools that don't support assertions. See
// prim_assert.sv for documentation for each of the macros.

`define ASSERT_I(__name, __prop)
`define ASSERT_INIT(__name, __prop)
`define ASSERT_INIT_NET(__name, __prop)
`define ASSERT_FINAL(__name, __prop)
`define ASSERT_AT_RESET(__name, __prop, __rst = `ASSERT_DEFAULT_RST)
`define ASSERT_AT_RESET_AND_FINAL(__name, __prop, __rst = `ASSERT_DEFAULT_RST)
`define ASSERT(__name, __prop, __clk = `ASSERT_DEFAULT_CLK, __rst = `ASSERT_DEFAULT_RST)
`define ASSERT_NEVER(__name, __prop, __clk = `ASSERT_DEFAULT_CLK, __rst = `ASSERT_DEFAULT_RST)
`define ASSERT_KNOWN(__name, __sig, __clk = `ASSERT_DEFAULT_CLK, __rst = `ASSERT_DEFAULT_RST)
`define COVER(__name, __prop, __clk = `ASSERT_DEFAULT_CLK, __rst = `ASSERT_DEFAULT_RST)
`define ASSUME(__name, __prop, __clk = `ASSERT_DEFAULT_CLK, __rst = `ASSERT_DEFAULT_RST)
`define ASSUME_I(__name, __prop)
```


RTL wasn't ready for ASIC tools

Challenges

- Initial blocks were used.

```
// Make sure we initialize the BRAM with the correct register reset value.  
/*  
initial begin  
    for (int k = 0; k < NUM_WORDS; k++) begin  
        mem[k] = WordZeroVal;  
    end  
end  
*/
```

RTL wasn't ready for ASIC tools

Challenges

- System functions were used in RTL.

```
function automatic integer vbits(integer value);  
`ifdef XCELIUM  
    // The use of system functions was not allowed here in Verilog-2001, but is  
    // valid since (System)Verilog-2005, which is also when $clog2() first  
    // appeared.  
    // Xcelium < 19.10 does not yet support the use of $clog2() here, fall back  
    // to an implementation without a system function. Remove this workaround  
    // if we require a newer Xcelium version.  
    // See #2579 and #2597.  
    return (value == 1) ? 1 : _clog2(value);  
`else  
    return (value == 1) ? 1 : _clog2(value);  
`endif  
endfunction
```

RTL wasn't ready for ASIC tools

Challenges

- Signals declared and initialized at the same statement.

```
logic unused_scramble_inputs;  
assign unused_scramble_inputs = scramble_key_valid_i & (~scramble_key_i) & (~RndCnstIbexKey) &  
                                (~scramble_nonce_i) & (~RndCnstIbexNonce) & scramble_req_q &  
                                ic_scr_key_req & scramble_key_valid_d & scramble_req_d;
```

Cell count is very small

Challenges

- Unrealistic cell count in range of 7 ~ 28 cells.

Cell	Reference	Library	Area	Attributes
U5	SAEDHVT14_OR4_1	saed14hvt_ff0p88v125c	0.399600	
U6	SAEDHVT14_TIE0_V1_2	saed14hvt_ff0p88v125c	0.488400	
U7	SAEDHVT14_BUF_U_0P5	saed14hvt_ff0p88v125c	0.266400	
U8	SAEDHVT14_BUF_U_0P5	saed14hvt_ff0p88v125c	0.266400	
U9	SAEDHVT14_INV_1	saed14hvt_ff0p88v125c	0.177600	
U10	SAEDHVT14_TIE1_4	saed14hvt_ff0p88v125c	0.399600	
core_busy_q_reg[0]	SAEDHVT14_FDPRBQ_V2LP_1	saed14hvt_ff0p88v125c	1.065600	n
core_clock_gate_i	prim_clock_gating_0		0.976800	h, n
gen_regfile_latch.register	ibex_register_file_latch_0_00000020_0_0_0_00000000		1441.712391	h, n, p
u_fetch_enable_buf	prim_buf_Width4		0.000000	h, p
u_ibex_core			3744.873607	
ibex_core_0_00000000_00000004_0_00000000_00000028_0_2_0_0_0_0_00000020_00000016_00000040				b, h, n, p
u_rf_rdata_a_ecc_buf	prim_buf_s00000020_0		0.000000	h, p
u_rf_rdata_b_ecc_buf	prim_buf_s00000020_1		0.710400	h, p
Total 13 cells			5191.336798	
1				

Unable to resolve designs in the link phase

Challenges

- Could be the cause for the above issue. Worked fine after PDK.

```
Linking design 'ibex_top'
Using the following designs and libraries:
```

```
-----
* (42 designs)                /mnt/hgfs/0_GP/Synthesis/ibex_top.db, etc
saed14hvt_ff0p88v125c (library)
                               /mnt/hgfs/0_GP/PDK/Syn_PDK/saed_14_pdk-20250329T143815Z-001/saed_14_pdk/hvt_ibex/saed14hvt_ff0p88v125c.db
saed14hvt_ss0p72vm40c (library)
                               /mnt/hgfs/0_GP/PDK/Syn_PDK/saed_14_pdk-20250329T143815Z-001/saed_14_pdk/hvt_ibex/saed14hvt_ss0p72vm40c.db
saed14hvt_tt0p8v25c (library)
                               /mnt/hgfs/0_GP/PDK/Syn_PDK/saed_14_pdk-20250329T143815Z-001/saed_14_pdk/hvt_ibex/saed14hvt_tt0p8v25c.db
```

```
Warning: Unable to resolve reference 'ibex_csr_32_0_0_0' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_32_0_0_6' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_7_0_0_0' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_32_0_0_5' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_32_0_0_4' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_32_0_0_3' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_32_0_0_2' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_32_0_0_1' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
Warning: Unable to resolve reference 'ibex_csr_7_0_0_1' in 'ibex_cs_registers_0_00000001_0_0_0_0_00000000_00000028_0_00000000_00000004_0_0_2_0'. (LINK-5)
```

0

Setup violations

Challenges

- We first tried to put set_dont_use for cells violating the timing paths. It worked at the end but was iterative. Not a root-cause solution.
- Traced back the violating cells found out that there was a high fanout net so we inserted buffers to help the driver.
- Also inserted constraints on max transition of critical pins.

```
#####  
# MY_EDIT_#  
#####  
insert_buffer -inverter_pair u_ibex_core/cs_registers_i/clock_i SAEDHVT14_INV_S_4  
set_dont_touch [get_net u_ibex_core/cs_registers_i/*]  
set_max_transition 1.0 [get_pins u_ibex_core/cs_registers_i/*]
```

Unmatched points in Formality

Challenges

- Assign statement used to connect signal_a to signal_b which should be connected to ground instead.

```
7677   assign \crash_dump_o[exception_pc][0] = \instr_addr_o[1] ;
7678   assign \crash_dump_o[next_pc][0] = \instr_addr_o[1] ;
7679   assign \crash_dump_o[current_pc][0] = \instr_addr_o[1] ;
```

3 Failing compare points (0 matched, 3 unmatched):

Ref	Port	r:/WORK/ibex_top/crash_dump_o\[current_pc][0]
Impl	None	

Ref	Port	r:/WORK/ibex_top/crash_dump_o\[exception_pc][0]
Impl	None	

Ref	Port	r:/WORK/ibex_top/crash_dump_o\[next_pc][0]
Impl	None	

Synthesis Results

- QoR of small Ibex
- QoR of max performance Ibex

QoR of small Ibex

Synthesis Results

Timing Path Group 'clk_i'

Levels of Logic:	9.00
Critical Path Length:	0.26
Critical Path Slack:	2.67
Critical Path Clk Period:	6.25
Total Negative Slack:	0.00
No. of Violating Paths:	0.00
Worst Hold Violation:	-0.02
Total Hold Violation:	-0.02
No. of Hold Violations:	1.00

Timing Path Group 'cluster_clock_gating'

Levels of Logic:	2.00
Critical Path Length:	0.17
Critical Path Slack:	0.00
Critical Path Clk Period:	6.25
Total Negative Slack:	0.00
No. of Violating Paths:	0.00
Worst Hold Violation:	-159.54
Total Hold Violation:	-36389.46
No. of Hold Violations:	1017.00

QoR of small Ibex

Synthesis Results

Cell Count

Hierarchical Cell Count:	1
Hierarchical Port Count:	118
Leaf Cell Count:	11354
Buf/Inv Cell Count:	1731
Buf Cell Count:	588
Inv Cell Count:	1156
CT Buf/Inv Cell Count:	0
Combinational Cell Count:	9341
Sequential Cell Count:	2013
Macro Count:	0

Area

Combinational Area:	3146.983216
Noncombinational Area:	1655.320794
Buf/Inv Area:	377.000400
Total Buffer Area:	173.52
Total Inverter Area:	218.49
Macro/Black Box Area:	0.000000
Net Area:	7081.797778
Cell Area:	4802.304010
Design Area:	11884.101788

QoR of max performance Ibex

Synthesis Results

Timing Path Group 'clk_i'

Levels of Logic:	6.00
Critical Path Length:	0.19
Critical Path Slack:	2.74
Critical Path Clk Period:	6.25
Total Negative Slack:	0.00
No. of Violating Paths:	0.00
Worst Hold Violation:	-0.04
Total Hold Violation:	-0.07
No. of Hold Violations:	2.00

Timing Path Group 'cluster_clock_gating'

Levels of Logic:	50.00
Critical Path Length:	5.85
Critical Path Slack:	0.20
Critical Path Clk Period:	6.25
Total Negative Slack:	0.00
No. of Violating Paths:	0.00
Worst Hold Violation:	-482.56
Total Hold Violation:	-189975.30
No. of Hold Violations:	2020.00

QoR of max performance Ibex

Synthesis Results

Cell Count

Hierarchical Cell Count:	19
Hierarchical Port Count:	1871
Leaf Cell Count:	14907
Buf/Inv Cell Count:	2531
Buf Cell Count:	743
Inv Cell Count:	1807
CT Buf/Inv Cell Count:	2
Combinational Cell Count:	12887
Sequential Cell Count:	2020
Macro Count:	0

Area

Combinational Area:	4478.583587
Noncombinational Area:	2105.892058
Buf/Inv Area:	537.506400
Total Buffer Area:	215.56
Total Inverter Area:	343.88
Macro/Black Box Area:	0.000000
Net Area:	9370.003317
Cell Area:	6584.475645
Design Area:	15954.478962

Formality Results

- Reports of small Ibex
- Reports of max performance Ibex

Reports of small Ibex

Formality Results

Reference design: r:/WORK/ibex_top
Implementation design: Imp:/WORK/ibex_top
2289 Passing compare points

Matched Compare Points	BBPin	Loop	BBNet	Cut	Port	DFF	LAT	TOTAL
Passing (equivalent)	0	0	0	0	276	988	1025	2289
Failing (not equivalent)	0	0	0	0	0	0	0	0

Reports of small Ibex

Formality Results

Reference design: r:/WORK/ibex_top
Implementation design: Imp:/WORK/ibex_top
2296 Passing compare points

Matched Compare Points	BBPin	Loop	BBNet	Cut	Port	DFF	LAT	TOTAL
Passing (equivalent)	0	0	0	0	276	2019	1	2296
Failing (not equivalent)	0	0	0	0	0	0	0	0

Effective Commands

`set_dc_allow_rtl_pg true`

`set_hdlin_keep_power_ground_pins true`

`set_verilogout_no_tri true`

`set_verilogout_equation false`

`set_fix_multiple_port_nets -all -buffer_constants`

`report_delay_calculation`

DFT in progress

Setup violates after DFT and we notice 2 high-fanout net

```
0:03:33 8956.5 8805.70 7934338.0 185.0 2737629.5000
0:03:34 8953.2 8805.70 7933724.0 185.0 2733956.7500
0:03:34 8952.2 8805.75 7933724.5 185.0 u_ibex_core/id_stage_i/imd_val_q_reg[0][29]/D 2732648.7500
0:03:35 8952.8 8805.75 7933724.5 185.0 u_ibex_core/wb_stage_i/g_writeback_stage.rf_wdata_wb_q_reg[29]/D 2732445.7500
0:03:36 8953.3 8805.72 7933723.5 185.0 u_ibex_core/id_stage_i/imd_val_q_reg[0][29]/D 2732595.0000
0:03:37 8956.1 8805.64 7933722.0 185.0 u_ibex_core/wb_stage_i/g_writeback_stage.rf_wdata_wb_q_reg[26]/D 2734403.0000
Information: State dependent leakage is now switched from off to on.
Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)
0:03:42 8957.2 8805.64 7933722.0 185.0 2740424.0000
Loading db file '/mnt/hgfs/0_GP/PDK/saed14_pdk/SAED14nm_EDK_CORE_HVT_v_062020/stdcell_hvt/db_ccs/saed14hvt_tt0p8v25c.db'
Loading db file '/mnt/hgfs/0_GP/PDK/saed14_pdk/SAED14nm_EDK_CORE_HVT_v_062020/stdcell_hvt/db_ccs/saed14hvt_ss0p72vm40c.db'
Loading db file '/mnt/hgfs/0_GP/PDK/saed14_pdk/SAED14nm_EDK_CORE_HVT_v_062020/stdcell_hvt/db_ccs/saed14hvt_ff0p88v125c.db'
```

Note: Symbol # after min delay cost means estimated hold TNS across all active scenarios

Optimization Complete

Warning: Design 'ibex_top' contains 2 high-fanout nets. A fanout number of 1000 will be used for delay calculations involving these nets. (TIM-134)
Net 'clk': 1589 load(s), 1 driver(s)
Net 'gen_regfile_ff.register_file_i/n992': 1984 load(s), 1 driver(s)

Inquiries

- Why there are negative increment values on the timing reports? We used `report_delay_calculation` and found that it add a negative and large value to the interpolation equation, however, the input transition and output capacitance are in the range specified by the LUT of the .lib file.

Inquiries

```
203 U8086/X (SAEDHVT14_INV_0P5) -0.04 3.57 f
```

```
Report : delay_calculation
Design : ibex_top
Version: 0-2018.06-SP1
Date   : Tue Apr 29 19:10:12 2025
*****

From pin:          U8086/A
To pin:            U8086/X
Main Library Units: 1ns 0.001pF 1000kOhm
```

```
Rise Delay

cell delay = 0.01805
Table is indexed by
  (X) input_pin_transition = 0.0503952
  (Y) output_net_total_cap = 0.862592
Relevant portion of lookup table:
      (X) 0.0292 (X) 0.0536
(Y) 0.5521 (Z) 0.0124 (Z) 0.0161
(Y) 1.9640 (Z) 0.0213 (Z) 0.0275

Z = A + B*X + C*Y + D*X*Y
A = 0.0055 B = 0.1143
C = 0.0043 D = 0.0706

Z = 0.01805
scaling result for operating conditions
multiplying by 1 gives 0.01805

Fall Delay

cell delay = -0.0489304
Table is indexed by
  (X) input_pin_transition = 0.475297
  (Y) output_net_total_cap = 0.784427
Relevant portion of lookup table:
      (X) 0.0879 (X) 0.1330
(Y) 0.5521 (Z) 0.0236 (Z) 0.0177
(Y) 1.9640 (Z) 0.0399 (Z) 0.0164

Z = A + B*X + C*Y + D*X*Y
A = 0.0152 B = 0.0230
C = 0.0359 D = -0.2768

Z = -0.0489304
scaling result for operating conditions
multiplying by 1 gives -0.0489304
```

Inquiries

- Are the given constraints meant to lbex with small architecture?
- Shall you provide the floorplan constraints?

Thank you